Moment Generating Function (MGF)

Statistics Reference Notes

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1. Moment Generating Function (MGF)

Definition

The **Moment Generating Function (MGF)** of a random variable X is defined as:

$$M_X(t) = E(e^{tX})$$

where

- E = expectation
- t = real parameter
- X = random variable

The MGF is called a **moment generating function** because it can be used to obtain the **moments of a distribution**, such as:

- Mean
- Variance
- Higher-order moments

Why MGF is Important

The derivatives of the MGF help compute statistical moments.

Mean (First Moment)

$$E(X) = M'_X(0)$$

Variance

$$Var(X) = M''_X(0) - [M'_X(0)]^2$$

Thus, the MGF provides a convenient method to compute **mean and variance** of probability distributions.

Concept Diagram

```
Random Variable X
        |
        v
Apply Expectation E(e^(tX))
        |
```

```
        v
Moment Generating Function Mx(t)
        |
        v
Moments of Distribution
(mean, variance, etc.)
```

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2. MGF for Continuous Random Variables

If X is a **continuous random variable** with probability density function f(x), the MGF is:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

Interpretation

1. Multiply e^{tx} with the density function f(x)
2. Integrate over the entire range of x

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3. MGF for Discrete Random Variables

If X is **discrete** with probability mass function P(x), then:

$$M_X(t) = \Sigma\, e^{tx} P(x)$$

Steps

1. Multiply e^{tx} with probability P(x)
2. Sum over all possible values of x

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4. Example of MGF

Consider a random variable:

X	P(X)
0	0.5
1	0.5

Step 1: Use formula

$$M_X(t) = \Sigma\, e^{tx} P(x)$$

Step 2: Substitute values

$$M_X(t) = e^{t(0)}(0.5) + e^{t(1)}(0.5)$$

Step 3: Simplify

$$M_X(t) = 0.5e^0 + 0.5e^t$$

Since $e^0 = 1$:

$$M_X(t) = 0.5 + 0.5e^t$$

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5. MGF of Binomial Distribution

If X ~ Bin(n, p), then the moment generating function is:

$$M_X(t) = (q + pe^t)^n$$

```
p = probability of success
q = 1 − p
```

Example

A coin is tossed **3 times**.

```
n = 3
p = 0.5
q = 0.5
```

MGF:

$$M_X(t) = (0.5 + 0.5e^t)^3$$

Sample Problem 1

Find the **MGF of a binomial distribution with n = 4 and p = 0.3**.

Solution

Given:

```
n = 4
p = 0.3
q = 0.7
```

Formula:

$$M_X(t) = (q + pe^t)^n$$

Substitute values:

$$M_X(t) = (0.7 + 0.3e^t)^4$$

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6. MGF of Poisson Distribution

If X ~ Poisson(λ), then:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

```
λ = average number of occurrences
```

Example

Suppose average number of accidents per day is:

```
λ = 2
```

MGF:

$$M_X(t) = e^{2(e^t - 1)}$$

Sample Problem 2

Find the **MGF of Poisson distribution when λ = 5**.

Solution

Formula:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

Substitute:

$$M_X(t) = e^{5(e^t - 1)}$$

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7. MGF of Geometric Distribution

If X follows a **geometric distribution**, its MGF is:

$$M_X(t) = pe^t / (1 - qe^t)$$

```
p = probability of success
q = 1 − p
```

Example

Let p = 0.4, q = 0.6:

$$M_X(t) = 0.4e^t / (1 - 0.6e^t)$$

Sample Problem 3

Find the **MGF of geometric distribution when p = 0.2**.

Solution

Given:

```
p = 0.2
q = 0.8
```

Formula:

$$M_X(t) = pe^t / (1 - qe^t)$$

Substitute:

$$M_X(t) = 0.2e^t / (1 - 0.8e^t)$$

Quick Comparison of MGFs

Distribution	MGF
Binomial	$(q + pe^t)^n$
Poisson	$e^{\lambda(e^t - 1)}$
Geometric	$pe^t / (1 - qe^t)$

Basic Level Summary

Topics Covered

- ✓ Moment Generating Function
- ✓ MGF definition
- ✓ MGF for discrete variables
- ✓ MGF for continuous variables
- ✓ MGF of Binomial distribution
- ✓ MGF of Poisson distribution
- ✓ MGF of Geometric distribution